CS & DA

Probability and Statistics



Permutations and Combinations

Lecture No. 06



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Recap of previous lecture



Topic

CIRCULAR PERMUTATION

Topics to be Covered



Topic P & C (Part 6)

1. Distribution of Different items
2. Distribution of Identical items

Q: How many 9 digit Nos can be formed using the digits 5,5, 6,6,6, 7,7,7,7

Sol: Total 9 digit Nos (RNA) $= \frac{n!}{p!q!r!} = \frac{9!}{2!\,3!\,4!}$

$p=2$, $q=3$, $r=4$, $n=9$ digits

Q: How many 9 digit Nos can be formed using the digits 2,2, 3,3,3, 4,4,4,4

Sol: Req Ans $= \frac{9!}{2!\,3!\,4!} = \frac{{}^9P_9}{2!\,3!\,4!}$

HW8 How many 4 digit Nos (≥6000) can be formed using 5,5,6,6,6,7,7,7,7

Sol: Total 4 digit Nos (≥6000) = 2 ways × 3 ways × 3 ways × 3 ways = 54 Nos
(RHA)

P_1	P_2	P_3	P_4
(6 or 7)	(5 or 6 or 7)	(5 or 6 or 7)	(5 or 6 or 7)

(Wild Guess)

It includes some wrong Nos as follows;

(6) 5 5 5 → ✗
(6) 6 6 6 → ✗
(7) 5 5 5 → ✗

∴ Correct Ans = 54 − 3 = 51 Nos

5 5 6 6 6 7 7 7 7

(M-II)

Various Nos:

6556
6557
6655
6657
6675
6755
6756
6765
⋮
7777

51 NOS

Distribution of Different items → No formula exist, only use Multiplication Rule.

Eg In how many ways (10 different) pencils can be distributed among (4 children) if

(1) there is No Restriction = ? (RA) $= \frac{4}{P_1}\times\frac{4}{P_2}\times\frac{4}{P_3}\times\frac{4}{P_4}\times\frac{4}{P_5}\times\frac{4}{P_6}\times\frac{4}{P_7}\times\frac{4}{P_8}\times\frac{4}{P_9}\times\frac{4}{P_{10}} = 4^{10}$ (Man Am)

WRONG APP.: Req Am $= \frac{10}{C_1}\times\frac{10}{C_2}\times\frac{10}{C_3}\times\frac{10}{C_4} = 10^4$ (Here children are picking pencils)

(2) one child gets 3 particular Pencils $= ({}^4C_1 \times {}^3C_3 \times 1) \times \frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{} = ({}^4C_1\times 1\times 1)\times 3^7$

(3) 1st " " 3 Particular " $= ({}^1C_1 \times {}^3C_3 \times 1) \times \frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{} = (1\times 1\times 1)\times 3^7$

(4) 1st " " exactly 3 pencils $= ({}^1C_1 \times {}^{10}C_3 \times 1) \times \frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{}\times\frac{3}{} = (1\times {}^{10}C_3\times 1)\times 3^7$

(5) one child gets exactly 3 pencils = ? Sensless Quest.

Note ① In how many ways 7 different pencils can be distributed among 3 children

$$\text{Req Ans (RA)} = \underline{3} \times \underline{3} \times \underline{3} \times \underline{3} \times \underline{3} \times \underline{3} \times \underline{3} = 3^7 \text{ ways}$$
$$P_1 \; P_2 \; P_3 \; P_4 \; P_5 \; P_6 \; P_7$$

② ~~one child get exactly 3 pencils~~ $= {}^4C_1 \times {}^{10}C_3 \times 1) \times 3^7$ ways

Blunder

$$= ({}^4C_1 \times {}^{10}C_3 \times 1) \times \underline{3} \times \underline{3} \times \underline{3} \times \underline{3} \times \underline{3} \times \underline{3} \times \underline{3}$$
$$(k_4) \qquad (k_4)(k_4)$$
$$P_1 \; P_2 \; P_3 \; P_4 \; P_5 \; P_6 \; P_7$$

there may exist such types of possibilities So we should avoid making questions.

Q In how many ways 3 different prizes can be awarded among 4 Boys if

① A Boy may get any Number of prizes = ? (RA) $= \frac{4}{P_1} \times \frac{4}{P_2} \times \frac{4}{P_3} = 64$ ways.

② No Boy gets more than one prize = ? (RNA) $= \underline{4} \times \underline{3} \times \underline{2} = 24$ ways

③ No Boy gets all the prizes = ? $= 64 - 4 = 60$ ways

Various possibilities when all the prizes are won by same student :

$\left(\frac{B_1}{P_1}\frac{B_1}{P_2}\frac{B_1}{P_3}\right)$ or $\left(\frac{B_2}{P_1}\frac{B_2}{P_2}\frac{B_2}{P_3}\right)$ or $\left(\frac{B_3}{P_1}\frac{B_3}{P_2}\frac{B_3}{P_3}\right)$ or $\left(\frac{B_4}{P_1}\frac{B_4}{P_2}\frac{B_4}{P_3}\right)$

Q: In how many ways B'day of 6 different persons will fall in Calendar months?

Sol: 6 Persons & 12 months

Total ways of falling B'day = ? = $\underset{P_1}{12} \times \underset{P_2}{12} \times \underset{P_3}{12} \times \underset{P_4}{12} \times \underset{P_5}{12} \times \underset{P_6}{12} = 12^6$ ways

No Restriction / (RA)

(ii) In how many ways B'days of 6 different persons will fall in exactly 2 Calendar months?

HW8

Ans = 4092

↳ WRONG App: = $\underset{J}{6} \times \underset{F}{6} \times \underset{M}{6} \times \underset{A}{6} \times \underset{M}{6} \times \underset{J}{6} \times \underset{J}{6} \times \underset{A}{6} \times \underset{S}{6} \times \underset{O}{6} \times \underset{N}{6} \times \underset{D}{6} = 6^{12}$

Distributions of Identical Items

If n identical items are to be distributed among r persons then possible Number of Distributions is

(1) Blank groups are allowed (BA) $= {}^{n+r-1}C_{r-1}$ (formula starts from zero)

(2) Blank groups are Not allowed (BNA) $= {}^{n-1}C_{r-1}$ (formula starts from one)

Eg if n = 4 identical Balls are to be distributed among 2 Children then possible No. of Distributions when

① Blank groups are not allowed = ? eg (2,2), (3,1), (1,3) i.e three ways

② " " are allowed = ? eg (4,0), (0,4), (2,2), (3,1), (1,3) i.e FIVE ways

Eg If n = 4 Identical Balls & r = 3 Children then possible Distributions when

① Blank groups are not allowed = ? eg (2,1,1), (1,2,1), (1,1,2) i.e three ways

② Blank groups are allowed = ? (2,1,1), (1,2,1), (1,1,2), (3,1,0), (3,0,1), (0,1,3), (0,3,1), (1,0,3), (1,3,0), (4,0,0), (0,4,0), (0,0,4), (2,2,0), (2,0,2), (0,2,2) i.e 15 ways

① $BNA = {}^{4-1}C_{3-1} = {}^{3}C_{2} = 3$ ② $BA = {}^{4+3-1}C_{3-1} = {}^{6}C_{2} = 15$

Q. In how many ways 10 identical pencils can be distributed among 4 children if

Sol: $n = 10, r = 4$

(1) A child may get any number of pencils = ? $= {}^{10+4-1}C_{4-1} = {}^{13}C_3$

(BA)

(2) Each child gets one or more pencils = ? $= {}^{10-1}C_{4-1} = {}^{9}C_3$

BNA

(3) Each child gets at least one pencils = ? $= {}^{9}C_3$

BNA

(4) A child may get nothing = ? $= {}^{13}C_3$

BA

(5) Each child gets something = ? $= {}^{9}C_3$

BNA

Q: In how many ways 16 Mangoes can be distributed among 3 Beggars if each beggar gets at least 3 Mangoes?

WRONG APP.: $n=16, r=3$, $BNA = {}^{16-1}C_{3-1} = {}^{15}C_2$

Correct App.: first we will distribute 3-3 Mangoes to each beggar and this can be done by one way. (According to condition)

Now Remaining Mangoes are $n=16-9=7$, $r=3$ Beggars & there is No Restriction for these 7 Mangoes so.

$$BA = {}^{7+3-1}C_{3-1} = {}^{9}C_2 = \frac{9\times 8}{2\times 1} = 36 \text{ ways.}$$

Hence Req ways $= 1 \times 36 = 36$ ways.

M-II $x+y+z=16$; $x\geq 3, y\geq 3, z\geq 3$

$x+y+z = 7+9$

$(x+y+z)-9 = 7$

$(x-3)+(y-3)+(z-3) = 7$

Let $x-3=u$, $y-3=v$, $z-3=w$

$\Rightarrow u+v+w=7$, $(x-3)\geq 0, (y-3)\geq 0, (z-3)\geq 0$

or $u+v+w=7$, $u\geq 0, v\geq 0, w\geq 0$

$n=7, r=3$, $BA = {}^{7+3-1}C_{3-1} = {}^{9}C_{2} = 36$

Note: Here in this Quest. Mangoes are not mentioned as Identical Mangoes, still we are assuming them as identical Mangoes bcoz we are performing NOBLE Job

Q8 How many Non-ve integral solutions are there of $x+y+z+w=100$

Note: set of true integers $= N = \{1,2,3,4\ldots\}$, set of Non-ve Integers $= W = \{0,1,2,3\ldots\}$

ATQ, $x+y+z+w=100$; $x, y, z, w \geq 0$

$n = 100$ Mangoes, $r = 4$ Beggars, BA

So BA $= {}^{100+4-1}C_{4-1} = {}^{103}C_3$

Analysis: Various Distributions: BA = 103

$(x,y,z,w) = (50, 40, 6, 4)$

$(25, 14, 1, 60)$

$(50, 0, 50, 0)$

$(0, 29, 21, 50)$

$\vdots$

$(100, 0, 0, 0)$

$(0, 100, 0, 0)$

$(0, 0, 100, 0)$

$(0, 0, 0, 100)$

$(0, 0, 0, 0)$ → ??

Am = 103

Note In last case

$x+y+z+w \neq 100$

so it is not possible

i.e Blanks are allowed But all Blanks are not allowed just beczg of SECURITY REASONS

(128) How many 6 digit Nos can be formed using the digits 4, 5, 6, 7 if

(I) Each Number has only one Triplet & Remaining digits are different = ? $= {}^4C_1 \times {}^3C_3 \times \frac{6!}{3!}$

eg (444567) or (555467) or (666457) or (777456) $= 4 \times \frac{6!}{3!}$

$= 6!/3!$ $\quad 6!/3!$ $\quad 6!/3!$ $\quad 6!/3!$

(II) Each Number has exactly 2 Triplets = ? $= {}^4C_2 \times \frac{6!}{3!3!} = 6 \times \frac{6!}{3!3!}$

eg (444555) or (444666) or (444777) or (555666) or (555777) or (666777)

(III) Each Number has exactly 3 pair of digits = ? $= {}^4C_3 \times \frac{6!}{2!2!2!}$

eg (445566) or (445577) or (556677) or (446677)

given digits = 4, 5, 6, 7

(4) Each Number has exactly one Triplet, one pair & one different digit = ?

eg (444556)

eg (454465)

$$= {}^{4}C_1 \times {}^{3}C_1 \times {}^{2}C_1 \times \frac{6!}{3!\,2!}$$

For Triplet — Doublet — Diff digit

(5) Each No. has exactly 2 pair of digits & Remaining digits are different = ?

eg (445567)

$$= {}^{4}C_2 \times {}^{2}C_2 \times \frac{6!}{2!\,2!} = 6 \times \frac{6!}{2!\,2!}$$

Pair — Diff digits

eg (445567) or (446657) or (447756)
(556647) or (557746) or (667745)

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THANK - YOU